PW SKILLS

ADVANCED EXCEL FUNCTIONS

VLOOKUP & STATISTICAL FUNCTIONS



Dataset Link: [Hospital Patient Dataset]

Q1. Retrieve the **Department** for Patient ID **P150** using a lookup formula that ensures **exact match and handles missing values gracefully.**

Q2. Extract the **Treatment Cost** for Patient ID **P210** using a function that allows **default value if not found** (no errors should appear).

Q3. Identify the **Doctor assigned** to Patient ID **P275** using a method that works even if **column positions change dynamically.**

Q4. Return the **City** for Patient ID **P180** when the lookup column is placed **to the right of the return column.**

Q5. Calculate total **Treatment Cost** for patients belonging to **Cardiology**, ensuring the formula remains **dynamic (no hardcoding).**

Q6. Determine the **number of patients with Insurance = "Yes"**, but ensure the formula works even if the dataset expands.

Q7. Compute total **Treatment Cost for patients from Mumbai,** and modify the formula so that city input can be changed from a cell reference.

Q8. Calculate the

- **Median Treatment Cost** and explain why median might be preferred over average in this dataset.
- **Maximum Treatment Cost.**

Q9. Compute the **Standard Deviation of Days Admitted** and interpret whether patient stay durations are **consistent or highly variable.**

Q10. Using a single formula, calculate total **Treatment Cost** where:

- Insurance = "Yes"
- AND Department = "Neurology"

(No helper columns allowed, must use logical multiplication)

Q11. Construct a lookup formula that returns "**Patient Not Found**" when the ID is missing, while ensuring **other errors are not masked unnecessarily.**

Q12. Calculate **Cost per Day** for each patient while ensuring:

- Division errors are handled.
- Output remains clean for reporting

Q13. Explain the differences between **VLOOKUP, XLOOKUP, and INDEX + MATCH.**

Based on their features, discuss **which function should be used in different scenarios and justify your.**

Q14. Transform Treatment Cost values by **rounding them to nearest 1000**, ensuring financial reporting consistency.

Q15. Create a formula Column “Final Payable Amount” that applies:

- **10% discount** if Insurance = "Yes"
- **5% surcharge** if Insurance = "No"

Output should reflect **final payable amount**